

Santhosh Narayanan Baburaman

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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026

B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

Publication: Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

SKILLS

- **Data & Analytics:** Python, SQL, R, Pandas, statsmodels, ETL/ELT, Data Modeling, Data Quality Testing
- **BI & Platforms:** Power BI (DAX), Tableau, PostgreSQL, Snowflake, BigQuery, Redshift, dbt, Alteryx, Streamlit
- **Methods & Cloud:** Mixed-Effects Regression, Hypothesis Testing, Time Series, Anomaly Detection, GCP, Git

EXPERIENCE

Data Analyst Intern | Philadelphia Department of Aviation Jun 2026 – Aug 2026

- Created the department's first structured reporting and data-verification layer across 2 revenue streams (~30% of airport operating revenue), replacing ~60 hrs/month of manual review; presented results to senior management
- Developed a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of billing data spanning 4 fiscal years, replacing manual Excel review (~40 hrs/month) and surfacing anomalies the prior process missed
- Rebuilt ground transportation reporting on PropWorks SQL → Amazon Redshift into Power BI, correcting aggregation-order and duplicate-rate defects that had misstated fiscal-year totals (~20 hrs/month saved)
- Integrated both data models to compute per-mode passenger capture rates and a seasonal Z-score anomaly detector with peer-deviation and Benford's Law checks

IT Student Assistant | USC Office of the Provost Jun 2025 – Present

- Provide IT support across 15+ administrative departments, configuring and reimaging devices for new users

Founder & President | ISE Analytics Student Association (IASA), USC Feb 2025 – May 2026

- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals

AI Intern | SmartInternz May 2023 – Jul 2023

- Led a 3-person team building a CNN-LSTM model that generates captions for 8,000 Flickr8k images in Python

Azure Cloud Intern | SmartKnower Sep 2022 – Oct 2022

- Deployed and managed virtual machines and containerized applications on Microsoft Azure cloud infrastructure

PROJECTS

Sleep & Cognition Research | Student Researcher, Suen Lab, USC (Python, BigQuery) Sep 2026 – Present

- Built an 11-step cleaning pipeline for 6 years (2021–2026) of Fitbit sleep logs across 5 source tables in BigQuery: overlap resolution, DST correction, cross-source deduplication, and one main sleep per person-night
- Linked 5 cognitive tests (Woodcock-Johnson battery, Stop-and-Go Switch, Figure Identification) to sleep across 7 time windows, using mixed-effects regression, between/within-person decomposition, and FDR-corrected testing
- Identified a wave/practice-effect confound that produced a spurious sleep–cognition association; added time controls as a standard step across all analyses

Automated Sales Analytics Copilot (Python, PostgreSQL, Claude API, Streamlit)  GitHub

- Designed a PostgreSQL star schema with 5 KPI views tracking revenue, margin, and discount impact across categories and regions on a 10K-order, \$2.3M dataset
- Automated anomaly detection and ARIMA/Prophet forecasting across 4 years of data, surfacing that Furniture margins collapse to -22% under high discounts, with nightly Slack and email alerts via scheduled GitHub Actions
- Created a natural language chat UI letting business users query KPIs, run what-if scenarios, and generate ad-hoc reports without writing SQL

Consulting Firm Data Warehouse (dbt, Snowflake, SQL, Python)

- Constructed a multi-layer ELT pipeline in dbt on Snowflake, transforming 17 normalized source tables into 3 subject-area datamarts for a 4-person team
- Modeled 7 conformed dimensions including an SCD Type 2 dimension preserving salary and title history, plus a generated date spine for consistent time aggregation
- Authored 39 automated data-quality tests and reconciled design docs against the production schema, documenting 13 discrepancies for the team